7. Adding Fingerprint Authentication for your apps

To develop an Android Application for Fingerprint authentication, you need to have a device containing a touch sensor, which is configured to secure the device and enrol at least one fingerprint.

To use the, Fingerprint authentication in your app, you will have to request the **USE_FINGERPRINT** permission within the project manifest file. Within the Android Studio Project tool fire up the **AndroidManifest.xml file** to add the fingerprint request permission as shown below:

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   package="com.arsaviva.fingerprint">

   <uses-permission
      android:name="android.permission.USE_FINGERPRINT" />
```

Fingerprint authentication depends on two system services - Keyguard-Manager and the FingerprintManager. Inside the onCreate method in your Activity, you will have to define the references to these two services as follows:

```
package com.arsaviva.fingerprint;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.app.KeyguardManager;
import android.hardware.fingerprint.FingerprintManager;

public class FingerprintActivity extends AppCompatActivity {

   private FingerprintManager fingerprintManager;
   private KeyguardManager keyguardManager;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_fingerprint_demo);

      keyguardManager =
                (KeyguardManager) getSystemService(KEYGUARD_SERVICE);
```